

ROSEHEARTY, United Kingdom

WMO: 030940

Lat: 57.700N

Lon: 2.117W

Elev: 23

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 95-12

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.1	32.1	23.5	17.7	34.2	25.4	19.5	35.1	51.9	42.0	47.1	42.5	9.7	170	0.794

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	69.1	62.0	66.3	60.0	63.8	58.7	63.0	67.6	61.2	64.9	59.7	62.7	12.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.9	80.0	65.1	59.4	75.7	63.2	58.1	72.1	61.4	28.5	67.8	27.3	65.2	26.2	62.8	71.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 40.6	34.6	30.6	26.1	76.0	3.4	3.3	23.7	78.4	21.7	80.3	19.7	82.2	17.2	84.6		
(5)			24.2	66.9	3.6	2.3	21.7	68.6	19.6	69.9	17.5	71.2	14.9	72.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	40.7	41.1	42.6	45.7	48.7	53.5	57.1	57.3	55.1	50.4	45.0	40.8		
	DBStd	7.32	3.63	4.27	4.03	4.45	3.90	3.94	3.72	3.43	3.55	3.57	4.38	4.59		
	HDD50	1478	288	250	232	142	72	10	0	0	3	39	156	286		
	HDD65	6136	753	669	696	580	506	345	247	239	298	454	599	750		
	CDD50	824	0	1	2	13	31	115	221	227	156	50	7	1		
	CDD65	4	0	0	0	0	0	0	2	1	1	0	0	0		
	CDH74	8	0	0	0	0	0	0	5	2	1	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	14.9	18.4	17.0	16.1	13.6	12.0	11.6	11.7	12.1	13.8	17.3	17.5	17.3		
	PrecAvg	33.5	3.0	2.4	2.1	2.3	2.3	2.7	2.8	2.8	2.6	3.9	3.6	2.9		
(15) Precipitation	PrecMax	41.1	8.8	4.4	4.3	5.4	5.2	5.0	5.2	6.2	10.5	7.9	7.0	6.4		
	PrecMin	23.5	1.1	0.5	0.5	0.6	0.7	1.0	1.3	0.7	1.1	0.9	1.7	1.1		
	PrecStd	4.2	1.6	1.1	0.8	1.3	1.0	1.2	1.3	1.3	1.9	1.6	1.5	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	55.5	56.2	61.7	65.1	70.1	72.9	72.7	70.7	62.2	57.7	55.3		
		MCWB	49.8	50.5	50.5	55.4	57.3	62.6	63.2	64.1	62.9	57.3	53.1	51.5		
	2%	DB	50.3	52.8	52.7	57.8	61.1	65.5	69.0	68.4	65.4	59.1	54.9	51.4		
		MCWB	47.4	48.8	48.1	51.9	54.9	59.7	61.9	61.8	59.2	54.9	51.7	48.6		
	5%	DB	48.0	50.3	50.4	54.8	57.8	62.3	66.1	65.2	62.4	57.2	53.1	48.9		
		MCWB	45.5	46.9	46.7	50.2	52.7	57.5	60.1	59.8	57.5	54.2	50.5	46.3		
	10%	DB	46.1	47.5	48.6	52.5	54.9	59.8	63.3	62.7	60.5	55.6	51.5	47.3		
		MCWB	44.2	44.5	45.7	48.8	50.7	55.6	58.6	59.0	56.4	53.2	49.3	44.9		
	0.4%	WB	50.2	51.2	51.4	56.2	58.9	63.2	65.8	64.8	63.7	58.0	54.4	51.6		
		MCDB	52.7	54.6	54.8	60.8	63.6	68.5	70.8	70.6	69.1	60.7	56.2	54.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.8	49.1	49.1	52.9	55.6	60.3	63.0	62.8	60.6	56.1	52.5	48.8		
		MCDB	49.8	52.5	51.7	56.2	59.6	64.6	67.5	67.0	64.2	57.9	54.4	50.8		
	5%	WB	45.9	47.1	47.5	50.9	53.3	58.2	61.0	61.1	58.5	54.8	51.0	46.9		
		MCDB	47.5	49.8	49.7	54.2	56.9	61.6	64.9	64.0	61.1	56.4	52.7	48.7		
	10%	WB	44.4	45.0	45.9	49.0	51.5	56.3	59.2	59.5	57.2	53.7	49.6	45.1		
		MCDB	46.0	47.1	48.1	51.9	54.2	59.2	62.4	61.7	59.4	55.3	51.2	46.9		
	5% DB	MDBR	6.9	7.7	8.0	9.3	9.9	9.0	9.2	8.9	9.1	8.4	7.8	7.7		
		MCDBR	9.6	11.3	11.6	13.7	15.1	14.5	14.7	13.7	12.5	10.8	9.2	9.7		
		MCWBR	8.5	9.2	8.5	9.6	10.3	9.8	9.2	8.6	8.5	8.7	8.3	8.8		
		MCWBR	9.0	10.4	10.4	13.4	14.0	14.2	13.6	12.1	11.3	9.8	8.3	9.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	8.3	9.1	8.3	9.9	9.9	10.0	9.4	8.3	8.4	8.5	7.9	8.7		
	taub		0.313	0.338	0.361	0.384	0.376	0.376	0.382	0.370	0.355	0.342	0.318	0.297		
		taud	2.237	2.322	2.333	2.298	2.379	2.409	2.401	2.441	2.481	2.420	2.277	2.170		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		168	221	249	261	272	274	269	265	251	218	165	134		
		Edh at Noon	16	23	29	35	35	34	34	30	25	21	15	13		
	RadAvg		130	329	681	1137	1550	1523	1473	1173	797	413	170	87		
(45) All-Sky Solar Radiation	RadStd		16	32	66	105	84	122	111	78	68	34	14	10		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page